

Gamma belief functions and fuzzy sets and application to combining predictive models

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ABSTRACT

Extending classic finite frameworks to continuous settings, this paper proposes the concept of gamma belief functions and gamma fuzzy sets. It shows that both the combination of gamma belief functions and the intersection of gamma fuzzy sets remain within the gamma family, enabling their application in combining gamma probability judgments in decision making and gamma regression models in ensemble learning. Using both simulated and real datasets, and under both constant and varying dispersion assumptions, experimental results show that the combined gamma regression models closely approximate the reference models learned from the full datasets, aligning with the objectives of bootstrapping. Notably, when evaluated against ground truth, the combined models outperform both the individual component models and the reference models in predictive accuracy.

1. Introduction

Probability judgment aggregation is a longstanding challenge in decision science, statistical modeling, and applied machine learning. It involves the integration of probabilistic assessments from multiple sources to produce a coherent representation of uncertainty. The problem initially arose in group decision-making where the conflicting interests of different constituents are balanced through combination and has later gained extensive interests in decision analysis, where a decision maker integrates individual probability judgments to form a single distribution to be used as an input to a formal Bayesian analysis (Genest and Zidek, 1986). The problem is particularly vital in domains where predictive accuracy depends on synthesizing diverse expert knowledge, such as in medicine, meteorology, and policy forecasting (Baars and Mass, 2005; Pepe, 2003; Tetlock, 2005). Its central premise lies at bringing together the strengths of different experts and consolidating their cognitive or information diversity to improve predictive performance (Satopää et al., 2016; Tetlock and Gardner, 2015).

Combining probability distributions is also important to the development of ensemble algorithms, which have become a foundation in modern machine learning, wherein multiple base learners are combined to yield predictive performance superior to that of individual models. It aims to integrate diverse models, predictions, learning algorithms, data perspectives, or model-specific characteristics to produce more accurate and reliable outputs in both supervised and unsupervised learning tasks (Re and Valentini, 2012). Among the classical ensemble strategies, bootstrapping plays a central role by constructing diverse mod-

els through different data partitions (Jacobs, 1995), as exemplified by a study showing that combining predictions by multiple neural networks yields better results than any single network could produce (Rogova, 1994) as well as by meta analysis in social sciences, where researchers often needs to combine scientific findings on a variable of interest using different datasets to form a better understanding on a phenomenon (Smith and Glass, 1977). Ad hoc bootstrapping methods include majority voting, weighted averaging, and Naive Bayes aggregation. Objective ensemble methods remain an active area of research, particularly under constraints such as limited labeled data, class imbalance, high dimensionality, and the challenge of maintaining diversity among base models.

The popular approaches to combining probability judgments are linear and log-linear opinion pooling. The linear opinion pooling combines probabilities as a weighted average of individual judgments. It is intuitive, but it lacks calibration and performance. The log-linear pooling multiplies probability distributions into an aggregate. It performs better than linear pooling but lacks theoretical justifications because the multiplication of prior probability distributions does not correspond to any meaningful construct in probability theory. The Bayesian justifications (e.g., Morris, 1977; Winkler, 1968) twist aggregation into sequential Bayes revisions by disguising judgmental priors as sampling information, introducing a middleman as the beholder of the consensus, and requiring the knowledge of non-existent middleman as prior along with additional mathematical assumptions such as invariance to scale and invariance to shift (Morris, 1977) or invariance to location (Winkler, 1981). The axiomatic justifications (e.g., Bordley, 1982), on the other

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hand, assumes the existence of a supernatural rational middleman who follows a set of normative axioms, which may be self-contradictory or contradictory to empirical observations.

This paper proposes a new approach to probability aggregation using soft computing methods. Soft computing, including fuzzy logic, rough sets, and Dempster-Shafer theory, has unique advantages in addressing uncertainty, imprecision, and incomplete information. These methods are widely applied to tackle core issues in high-dimensional spaces, imbalanced datasets, and prediction uncertainty. By enhancing model diversity, robustness, and interpretability, soft computing contributes significantly to ensemble learning frameworks. In particular, fuzzy logic and Dempster-Shafer theory facilitate the development of highly adaptive and interpretable ensemble systems suitable for complex real-world applications. Dempster's rule of combination (Dempster, 1967; Shafer, 1976), with its solid mathematical foundation and reliability, offers a principled alternative to heuristic aggregation methods like majority voting or linear pooling. Evidential ensemble methods based on this rule assign belief masses to model outcomes according to their reliability, allowing for a nuanced and flexible approach to aggregating predictions.

This paper proves that Dempster's rule can arrive at the same conclusions as in the existing Bayesian justifications for log-linear pooling without their assumptions and theoretical stretches. It applies the soft computing approach to combining gamma probability judgments and predictive models. The gamma distribution, with its shape and scale flexibility, provides a natural framework for modeling skewed, positive-valued variables. Its applications span time-to-event modeling, financial risk assessment, and reliability analysis, among others. In practice, gamma distributions are often employed when data are strictly positive and exhibit asymmetry, such as insurance claims (Boland, 2007), rainfall amounts (Aksoy, 2000; Wilks, 1990), or equipment lifetimes (Vineyard et al., 1999). In Bayesian learning, gamma distributions serve as a conjugate prior for exponential distributions and are useful for modeling precision parameters of normal distributions. While Gaussian models dominate the foundations of statistical learning, gamma models are increasingly relevant for problems involving multiplicative effects and heteroscedasticity. In this context, combining gamma-distributed judgments or models becomes essential when constructing ensemble systems or performing meta-analytical integration. Toward this end, the paper proposes gamma belief functions and gamma fuzzy sets as extensions of respective classic knowledge representation tools for uncertainty and ambiguity and a novel approach to aggregating gamma probability judgments. Specifically, it defines the concept of gamma belief functions, thereby generalizing continuous belief functions beyond the Gaussian family (Dempster, 1990; Liu, 1996; Liu et al., 2006). It proposes the notion of gamma fuzzy sets, enabling the fusion of imprecise judgments within a fuzzy set framework. It shows that the intersection of gamma fuzzy sets and the combination of gamma belief functions both yield gamma structures, preserving closure and interpretability. The result leads to a new evidential method for gamma probability aggregation.

The paper demonstrates that the new evidential fusion mechanism applies not only to probability judgments but also to predictive models, particularly gamma regressions. Gamma regressions, which are generalized linear models with gamma-distributed responses, are suited for skewed continuous targets in predictive analytics. The paper shows how independent gamma regressions can be combined into a single model via belief function or fuzzy set operations. Although many prior studies apply Dempster's rule to ensemble learning, they all entail re-coding predictions as finite belief functions. The fusion method this paper proposes directly combines the predictions as continuous gamma belief functions or fuzzy sets with no need for re-coding or discretization, offering a computationally efficient and theoretically grounded alternative to data-level bootstrapping. Through simulations and applications to real-world data, our results show that the combined models approximate or even outperform the reference models built on fully merged datasets. This finding has implications for ensemble learning, distributed data analysis, and systematic evidence synthesis when data merging is impractical.

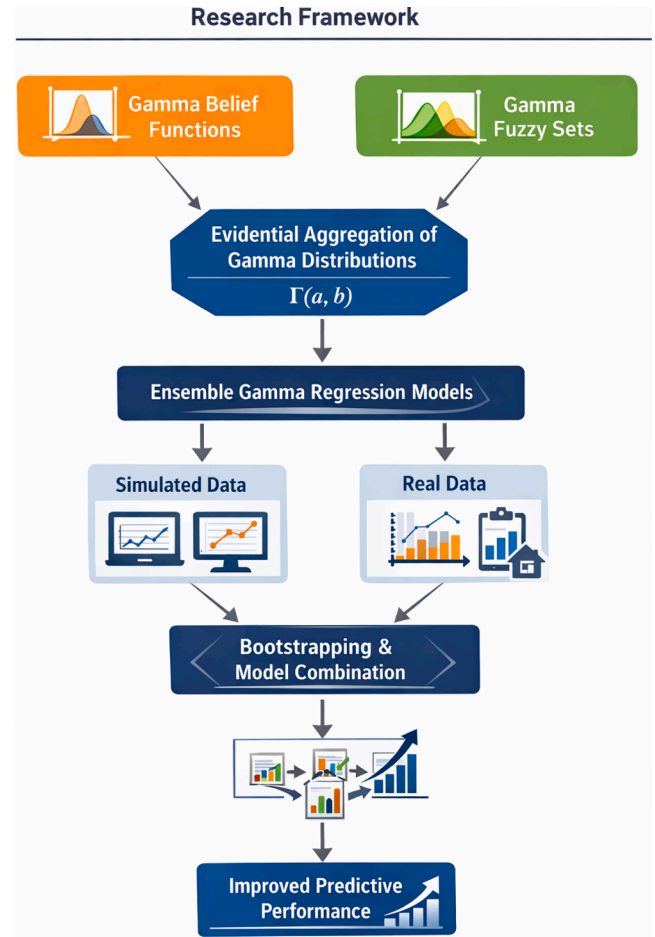


Fig. 1. Research framework.

The rest of the paper is structured as follows with the research framework displayed in Fig. 1. Section 2 reviews related work on probability aggregation and evidential ensemble learning. Section 3 reviews Bayesian and axiomatic approaches to probability aggregation, proposes the concepts of gamma belief function and gamma fuzzy sets, and applies Dempster's rule to combining gamma belief functions and the normalized product intersection rule to combine gamma fuzzy sets to establish a formula for evidential aggregation of gamma distributions. Section 4 applies the gamma probability aggregation formula to the evidential ensemble learning of both classic and generalized gamma regression models and evaluates the ensemble method using both synthetic and real datasets through experiments. Section 5 concludes the paper with a summary and a discussion.

2. Related studies

Three major traditions inform the study of probability aggregation: behavioral, structural, and functional approaches. Behavioral studies focus on how human cognitive biases and heuristics affect judgment aggregation, often highlighting the impact of linguistic framing or noise filtering capabilities. They investigate that, for examples, linguistic cues can significantly influence aggregated perceptions (Mislavsky and Gaertig, 2022), and the cognitive ability to filter out noise may play a more significant role in improving forecast accuracy than bias reduction or efficient information extraction (Satopää et al., 2021).

Structural approaches emphasize the connectedness of expert opinions in time and space, as in hierarchical or time-dependent models that update expert beliefs incrementally. Techniques like the Delphi method enable iterative consensus formation (Dalkey and Helmer, 1963), while

dynamic Bayesian structures accommodate sparse and asynchronous inputs. For examples, considering the variation of expert opinions across different organizational units, [Lipscomb et al. \(1998\)](#) proposed a hierarchical method to combine individual judgments as Gaussian distributions, and [Satopää et al. \(2014b\)](#) presents a dynamic hierarchical model for aggregating sparse expert beliefs over time, accommodating experts updating their beliefs at random intervals, producing sharp and well-calibrated aggregate probabilities. [Ivanovska and Slavkovik \(2025\)](#) explores methods for aggregating probabilistic opinions on logically related issues, extending classical judgment aggregation frameworks to handle probabilistic judgments. Along the same line, combining causal opinions becomes a new trend in extending probability aggregation to judgments of causal relations ([Alrajeh et al., 2020](#); [Bradley et al., 2014](#); [Friedenberg and Halpern, 2018](#)).

Functional approaches, which are central to this work, aim to establish mathematically grounded aggregation rules. The two most popular formulas are the Kalman filter and the linear opinion pooling. The former uses the multiplicative formula to combine information from two sources as Gaussian distributions ([Kalman, 1960](#)): the predictions based on the last known state of a system and the measurements of the observable as the manifestations of the current system state, to provide the best possible estimate of the current state of the system. The latter uses additive formula to combine expert probability assessments as the weighted sum of individual distributions with subjective weights selected by the decision maker based on performance, skill, or reliability ([Stone, 1962](#)). Another formula is geometric averaging. For example, assuming the log-odds follow a Gaussian distribution centered at a true log-odd adjusted by a factor reflecting system bias, [Satopää et al. \(2014a\)](#) proposed an aggregator as geometric mean of the odds.

The linear opinion pooling is ad hoc in nature and lacks calibration and sharpness ([Ranjan and Gneiting, 2010](#)) and performs poorly relative to multiplicative pooling ([Allard et al., 2012](#)). The multiplicative rule or its equivalent log-linear pooling is semantically appealing when the expert beliefs are based on private information ([Dietrich and List, 2017](#)). To justify or derive the multiplicative formula, however, probabilistic formations treat expert judgments as inputs to a posterior update, often under assumptions such as independence or conjugacy. For instance, when expert beliefs are modeled using beta distributions, they can be sequentially updated to produce a new beta distribution, reflecting combined evidence ([Winkler, 1968](#)). Gaussian beliefs have also been combined under such frameworks ([Morris, 1977](#); [Winkler, 1981](#)), with Kalman filtering emerging as a canonical example ([Kalman, 1960](#)).

There also exist axiomatic justifications for multiplicative pooling methods that attempt to derive aggregation rules from rationality postulates. For example, framing the combination of probability judgments into a multi-criterion decision problem where multiple criteria are aggregated into a multi-attribute utility function, [Bordley \(1982\)](#) arrives at a weighted multiplicative formula by assuming the decision maker's belief ordering follows a set of axioms similar to those for the preference ordering that leads to the expected utility theory. The weights may be further specified as either τ parameters measuring data dependencies among experts ([Krishnan, 2008](#)) or a single ν parameter summarizing all dependencies ([Polyakova and Journeel, 2007](#)).

Both Bayesian and axiomatic justifications has conceptual limitations and flaws. Bayesian justifications twist the combination of parallel expert judgments as sequential Bayesian revisions, with judgmental priors disguised as sampling information, a middleman introduced as the beholder of the consensus, and the knowledge of non-existent middleman required as prior. For example, the natural-conjugate method ([Winkler, 1968](#)) applies to a special case when a probability judgment is expressed as a natural-conjugate prior so that multiple judgments are combined via successive applications of Bayesian theorem into a posterior distribution. When expert judgments on the success probability in Bernoulli trials are expressed as beta priors, the combination is a posterior distribution, which is yet another beta distribution with the parameters as the sum of those in expert priors ([Winkler, 1968](#)). Taking a sim-

ilar Bayesian approach but casting expert judgments as likelihood functions to be combined with the prior judgment by the middleman, [Morris \(1977\)](#) arrives at the multiplicative formula for combining Gaussian distributional judgments, under invariance to scale and invariance to shift assumptions. In a similar attempt, [Winkler \(1981\)](#) considers the dependence of information sources under the location-invariance assumption to combine Gaussian distributions.

The axiomatic justification is normative in nature; it assumes the middleman should behave as a rational supernatural agent who follows the stipulated axioms. However, these axioms are either self-contradictory ([Schervish, 1986](#)) or contradictory to psychological observations. For example, the similar axioms for preferences have shown be consistently violated in experiments, leading to descriptive theories such as prospect theory, rank- and sign-dependent utility theories ([Kahneman and Tversky, 1979](#); [Liu, 2004](#); [Luce and Fishburn, 1991, 1995](#)), or their approximation, coarse utility theory ([Liu, 1999](#); [Liu and Shenoy, 1995](#)), as alternatives to expected utility theory. Recent work on axiomatic justification has focused on properties like consensus preservation, plausibility preservation, and diversity ([Dietrich and List, 2017](#)), and alternative aggregation mechanism based on information fusion and evidence theory have gained traction in information sciences.

This paper builds on the third tradition by proposing an interpretation of probability judgments as belief functions and fuzzy sets, thus avoiding some of the conceptual pitfalls of Bayesian revisions with fictitious intermediaries or hidden priors. Unlike prior studies restricted to discrete or Gaussian settings, we extend belief combination to gamma distributions and demonstrate their utility in regression modeling. Our approach aims to be prescriptive and intuitive, resembling the usual connotation of information fusion or evidence pooling. The combination of fuzzy sets ([Zadeh, 1965](#)) is merely an extension of set intersection to the cases when the involved sets are fuzzy. Dempster's rule ([Dempster, 1966, 1967](#)) for combining belief function is a derivative of probability theory applied to two probability spaces, source and target, connected by multivalued mappings, with the source space encoding evidence, the target space representing the uncertainties of interest, and the combination corresponding to the multiplication of probabilities of independent events in the source space. Conceptually, any probability distribution is a Bayesian belief function with focal elements are singletons or hyperplanes ([Shafer, 1976](#)), and thus the combination of probability judgments is naturally interpreted as the combination of independent evidence implying the judgments. Using this approach, this paper arrives at the same results as [Winkler \(1968\)](#) by combining beta distributions and as [Morris \(1977\)](#) by combining Gaussian belief functions without the middleman and his prior.

Belief functions are traditionally restricted to finite target space of interest ([Shafer, 1976](#)). An exception is Gaussian belief functions ([Dempster, 2001](#); [Liu, 1996](#); [Shafer, 1992](#)) that extend the concepts to a continuous frame. Gaussian belief functions exhibit the closure property; the combination via Dempster's rule is also Gaussian, and they include the Kalman filtering as a special case ([Dempster, 1990](#); [Liu, 2018](#)). There has been no study of any other family of distributions that may form a proper class of continuous belief functions except for a recent conference presentation on the notion of gamma belief functions ([Liu, 2024](#)). This paper attempts to push the envelop. It builds on the preliminary work and studies gamma belief functions in a broader context of aggregating probability judgments. It compares Bayesian, axiomatic, evidential, and fuzzy approaches to probability aggregation, proposes the concept of gamma fuzzy sets, and proves that the gamma probability judgments may be combined naturally as gamma belief functions via Dempster's rule or gamma fuzzy sets via the normalized product-intersection rule without the assumptions required by the Bayesian and axiomatic approaches.

This paper also proposed a new evidence-based ensemble learning method. The common feature of all evidential methods is to subjectively transform classifications or predictions on a finite variable of interest into discrete belief functions and aggregate them via Dempster's rule,

or its modifications, as support for the predictions. Their differences lie at how to code model predictions. Xu et al. (1992) uses the performance of classifiers such as recognition, substitution, and rejection rates as the basis for mass assignments to predictions. Rogova (1994) and Al-Ani and Deriche (2002) measure the evidence based on the distance between an output and its reference vector, which is the average outputs of the classifiers that make correct predictions on a class (Rogova, 1994) or its refined version by iteratively minimizing the error between the combined predictions and the actual target labels among training examples (Al-Ani and Deriche, 2002). Bi et al. (2008) uses the highest two predictions of each class as evidential support for the class to produce triplet mass assignments. Tabassian et al. (2012) addresses label ambiguity in training examples, constructs complementary feature spaces to create different classifiers, and interprets their outputs as mass assignments. Deng et al. (2016) uses information from the confusion matrix to improve the basic belief assignment. Liu et al. (2018) uses reliability to transform the model predictions to mass assignments. Qin and Xiao (2019) uses distance and similarity of interval numbers obtained from k-means++ clustering method (Arthur and Vassilvitskii, 2007) as evidence. Similar techniques in coding and combining evidence are also used in designing evidential classifiers. For example, Denoeux uses the distance between an object and its k-nearest neighbors (Denoeux, 1995), or the distance between a pattern and a reference pattern (Denoeux, 2000), or individual features (Denoeux, 2018, 2019) as evidence. In contrast, my method for combining gamma regressions deals with predictions on a continuous variable of interest. There is no need for re-interpretation or re-engineering of the model outputs, and instead, the model predictions are directly combined as continuous belief functions into combined predictions.

3. Combining gamma probability judgments

For a variable of interest, Y , assume there exist n judgments of its probability distribution, say, $f_i(y), i = 1, 2, \dots, n$, by a group of independent experts. An intuitive approach for a group judgment is the weighted sum or linear opinion pooling (Krogh and Vedelsby, 1994; Stone, 1962): $w_1 f_1(y) + w_2 f_2(y) + \dots + w_n f_n(y)$, with the weights being subjectively chosen by whoever has knowledge of each expert's judgmental reliability and capability. A second approach, based on Bayesian theory, arrives at the multiplicative formula $f_1(y)f_2(y) \dots f_n(y)$ or its weighted variation $f_1(y)^{w_1} f_2(y)^{w_2} \dots f_n(y)^{w_n}$ (Bordley, 1982; Morris, 1977; Winkler, 1968) or its equivalent logarithmic opinion pooling $w_1 \ln f_1(y) + w_2 \ln f_2(y) + \dots + w_n \ln f_n(y)$ (Heskes, 1997). Of course, since $f_1(y), f_2(y), \dots, f_n(y)$ are prior probabilities on the same variable, their multiplication does not correspond to any meaningful constructs in Bayesian theory, but researchers justified this multiplicative formula from Bayesian revision and rational decision perspectives.

Winkler (1968) proposes the natural-conjugate approach for the case when $f_i(y), i = 1, \dots, n$, are natural conjugate priors so that when combined with a likelihood function using Bayesian theorem, the posterior and the prior have an identical parametric form. For example, assume $f_i(y) = \text{beta}(y; \alpha_i, \beta_i)$ is a beta prior with parameters α_i and β_i by Expert i . After observing s successes in k Bernoulli trials, the expert's posterior belief about the success probability will be $\text{beta}(y; \alpha_i + s, \beta_i + k)$. Thus, assuming that Expert i 's prior is tantamount to sample information of observing α_i successes in β_i trials, Winkler combines $f_1(y), f_2(y), \dots, f_n(y)$ as successively applications of Bayesian theorem: combining Expert j 's prior judgment $\text{beta}(y; \alpha_j, \beta_j)$ with Expert i 's prior $\text{beta}(y; \alpha_i, \beta_i)$ as if combining prior $\text{beta}(y; \alpha_j, \beta_j)$ with binomial sampling function for observing α_i successes in β_i trials, resulting the posterior distribution $\text{beta}(y; \alpha_i + \alpha_j, \beta_i + \beta_j)$, which can be further treated equivalently as sample information of observing $\alpha_i + \alpha_j$ successes in $\beta_i + \beta_j$ trials to be combined with yet another expert's beta prior. Following this rationale, Winkler proposes combining beta priors $\text{beta}(y; \alpha_i, \beta_i), i = 1, 2, \dots, n$, as $f(y) = \text{beta}(y; \sum w_i \alpha_i, \sum w_i \beta_i)$ with weights w_i subjectively chosen per Expert i 's reliability.

The second perspective, also based on Bayesian theorem, takes the view point that a decision maker receives Expert i 's judgment $f_i(y)$ as an observation in an experiment, though such a judgment merely reflects the state of knowledge to the expert. Morris (1977) shows that, when an expert is calibrated in a Gaussian probability judgment $f_E(y; \mu, \sigma^2)$ on an unknown variable Y , then the posterior judgment of Y given the expert opinion is normalized product of the expert prior f_E and the decision maker's prior $f_D(y)$: $f_D(y|\mu, \sigma^2) = c f_E(y; \mu, \sigma^2) f_D(y)$, with c being a normalization constant. The assessment of the expert must satisfy the following two assumptions for the result to hold. The first is the invariance to variance, which implies $f_D(y|\sigma^2) = f_D(y)$, i.e., the variance of the expert's prior alone provides no information to the decision maker about the uncertain quantity. The second is the invariance to shift, which stipulates $p(\mu = \mu_0 | \sigma^2, y + \Delta) = p(\mu = \mu_0 - \Delta | \sigma^2, y)$ on the decision maker's likelihood assessment of the mean value of the expert's prior given the variance σ^2 and the revealed value y . When receiving opinions from multiple experts, Morris (1977) suggests summarizing the judgments $f_1(y), f_2(y), \dots, f_n(y)$ into a composite prior $h(y)$ as the opinion by a single representative expert. If the experts are independent and calibrated, then $h(y) = c f_1(y) f_2(y) \dots f_n(y)$.

Following the development of additive multi-attribute utility functions, the third approach is axiomatic (Bordley, 1982). Assume a group of n experts give probability judgments $p_i, i = 1, 2, \dots, n$ on an event or equivalently their odd ratios $o_i = \frac{p_i}{1-p_i}$. Define a belief ordering $\geq$ over the odd ratio compositions $(o_1, o_2, \dots, o_n)$ such that $(o_1, o_2, \dots, o_n) \geq (o'_1, o'_2, \dots, o'_n)$ if and only if the decision maker feels at least as sure about the event occurring when the experts assess odds $(o_1, o_2, \dots, o_n)$ as when they assess odds as $(o'_1, o'_2, \dots, o'_n)$. Then, there exist continuous functions $u_1, u_2, \dots, u_n$ such that $(o_1, o_2, \dots, o_n) \geq (o'_1, o'_2, \dots, o'_n)$ if and only if $\sum_{i=1}^n u_i(o_i) \geq \sum_{i=1}^n u_i(o'_i)$ if $n \geq 3$, the belief ordering is a continuous weak ordering satisfying the non-interactive property, and all odd ratios are positive real numbers. Here the non-interactive property is similar to preferential independence for preference ordering (Fishburn, 1970) and stipulates that if an expert shows no difference in two different scenarios, she can be ignored while deciding which scenario is more sure to occur. For example, Expert 1 is non-interactive with respect to others if there exists judgment o_1^* such that $(o_1^*, o_2, \dots, o_n) \geq (o_1^*, o'_2, \dots, o'_n)$, then for any judgment by this expert, $o_1^{**}, (o_1^{**}, o_2, \dots, o_n) \geq (o_1^{**}, o'_2, \dots, o'_n)$. With these axioms, the belief ordering over odds $(o_1, o_2, \dots, o_n)$ can be characterized by a monotonic function $o_G = G(\sum_{i=1}^n u_i(o_i))$ as the group judgment. Imposing further Bayesian conditions, Bordley (1982) derives $o_G = (o_0)^{1-\sum_{i=1}^n w_i} \prod_{i=1}^n (o_i)^{w_i}$ with o_0 as the prior odd. Plugging $o_i = \frac{p_i}{1-p_i}, i = 0, 1, 2, \dots, n$ and $o_G = \frac{p_G}{1-p_G}$, we will obtain p_G as multiplication of individual $p_i, i = 1, 2, \dots, n$. The third approach is normative in nature: if the decision maker is rational, she should follow the multiplicative formula to combine individual judgments. Of course, descriptively, people may violate one or more these axioms. For example, the counter part of non-interactivity, preferential independence, is shown to be violated in experiments as exemplified by the Allais paradox (Allais, 1953).

In the rest of this section, I propose combining probability judgments as belief functions using Dempster's rule as well as the intersection of fuzzy sets. We will apply the new approach to gamma probabilities, but as a side result, we will see that this approach includes Winkler's combination of beta distributions and Morris' combination of Gaussian distributions as special cases.

3.1. Dempster's rule of combination

Belief functions represent epistemic knowledge on the whereabouts of truth in a frame of discernment Θ , within which each subset represents a proposition that the truth lies in it. A central idea in its representation is limited divisibility of beliefs (Liu and Yager, 2008): a belief function is made of one or more focal elements A and probability mass numbers $m(A)$ assigned to the focal elements such that

$$\sum \{m(A) \mid A \subseteq \Theta\} = 1. \quad (1)$$

Focal elements are atomic, and mass assignments are indivisible; a mass number $m(A)$ allocated to focal element A cannot be cut into smaller proportions for reassignment to proper subsets of A . For example, full ignorance implies no knowledge of any form to locate the whereabouts of the truth in Θ . We represent it by a vacuous belief function: $m(\Theta) = 1$, where Θ is the only focal element, and any positive mass $m(A)$ for any proper subset A indicates the existence of knowledge and contradicts ignorance.

Dempster's rule combines belief functions that code independent items of evidence. Although Shafer (1976) adopts the rule as a definition, Dempster (1967) derived it using multivalued mappings between two probability spaces Ω and Θ : $\omega \rightarrow A$, such that probability mass m assigned to a single point $\omega \in \Omega$ in the source space is transferred to the target space as $m(A)$, measuring the support to proposition $A \subseteq \Theta$. Thus, both A and $m(A)$ are indivisible because breaking them up into smaller units is not faithful to the original evidence. Assuming there exist two independent pieces of evidence implying two separate multivalued mappings: $\omega_1 \rightarrow A_1$ and $\omega_2 \rightarrow A_2$, where $\omega_1, \omega_2 \in \Omega$ and $A_1, A_2 \subseteq \Theta$. Assume mass m_1 assigned to ω_1 is transferred to Θ as $m_1(A_1)$ and m_2 to ω_2 transferred as $m_2(A_2)$. Because of independence of evidence implies the independence of the probability distributions in Ω , and so the joint mass assigned to (ω_1, ω_2) is $m_1 m_2$, which is transferred to the target space as $m_1(A_1)m_2(A_2)$ as a partial support to the proposition $A_1 \cap A_2$, if it is not empty, and the intersection $A_1 \cap A_2$ becomes a new focal element for the combined belief function. The mass assignment for the combined evidence is then the following up to normalization:

$$m(A) \approx \sum \{m_1(A_1)m_2(A_2) \mid A_1 \cap A_2 = A \neq \phi\}. \quad (2)$$

An empty intersection indicates a conflict, and $\sum \{m_1(A_1)m_2(A_2) \mid A_1 \cap A_2 = \phi\}$ measures the degree of conflict.

Example 1. This example illustrates Dempster's rule. Assume we want to find the culprit, among three suspects, Tony (T), Smith (S), and Dick (D), who committed a bank burglary. Mrs. Johnson reported that she saw a burly guy near the bank around the time of the crime. Assume Mrs. Johnson's witness is 60% reliable based on her eye sight. If her testimony is reliable, the evidence pinpoints to Tony or Dick since they are broad-shouldered. Thus, $m_1(\{T, D\}) = 0.6$. However, if she is not reliable, the testimony carries no information, i.e., $m_1(\{T, S, D\}) = 0.4$. The surveillance camera recorded a fuzzy picture of the person's eyes, which are four times more likely to be black than gray. The second item of evidence suggests $m_2(\{S\}) = 0.8$ and $m_2(\{T, D\}) = 0.2$ since Smith has black eyes. Now let us combine the two pieces of evidence using Dempster's rule. First, since $\{T, D\} \cap \{S\} = \phi$, $m_1(\{T, D\}) \times m_2(\{S\}) = 0.48$ is considered the degree of conflict. Two possible hypotheses are respectively $\{S\} = \{T, S, D\} \cap \{S\}$ with support $m_1(\{T, S, D\}) \times m_2(\{S\}) = 0.32$ and $\{T, D\} = \{T, D\} \cap \{T, D\} = \{T, S, D\} \cap \{T, D\}$ with support $m_1(\{T, D\}) \times m_2(\{T, D\}) + m_1(\{T, S, D\}) \times m_2(\{T, D\}) = 0.12 + 0.08 = 0.2$. Therefore, the combined evidence points to Smith as the most likely culprit:

$$m(\{S\}) = \frac{0.32}{1-0.48} = 0.615,$$

$$m(\{T, D\}) = \frac{0.20}{1-0.48} = 0.385.$$

Frame Θ is assumed to be finite in the classic theory (Shafer, 1976), but currently there are two extensions to the continuous frames: Gaussian belief functions that assign Gaussian mass densities over parallel hyperplanes as focal elements (Dempster, 2001; Liu, 1996) and belief functions on real numbers that assign mass densities over closed intervals as focal elements (Smets, 2005). When Θ is continuous, basic mass assignments must be replaced by mass density functions. Assume Y is a continuous random variable and $f_1(y)$ and $f_2(y)$ are the mass density functions from two independent sources, then their combination is another mass density

$$f_1(y) \oplus f_2(y) = \frac{f_1(y)f_2(y)}{\int f_1(y)f_2(y)dy} \quad (3)$$

note here the multiplication of mass numbers is replaced by $f_1(y)f_2(y)$, and the sum of all the multiplications is replaced by the integral $\int f_1(y)f_2(y)dy$.

For example, let $f_i(y) = \text{beta}(y; \alpha_i, \beta_i)$, for $i = 1, 2$. It is easy to verify that

$$\text{beta}(y; \alpha_1, \beta_1) \oplus \text{beta}(y; \alpha_2, \beta_2) = \text{beta}(y; \alpha_1 + \alpha_2, \beta_1 + \beta_2) \quad (4)$$

by noting

$$\int_0^1 y^{\alpha_1+\alpha_2-1} (1-y)^{\beta_1+\beta_2-1} dx = \frac{\Gamma(\alpha_1 + \alpha_2)\Gamma(\beta_1 + \beta_2)}{\Gamma(\alpha_1 + \alpha_2 + \beta_1 + \beta_2)}$$

in applying Eq. (3). As we see, Eq. (4) produces the same result as the natural-conjugate method by Winkler (1968) without casting the combination as a data-generating process. Also, per Eq. (4), $\text{beta}(y; 0, 0)$ is a vacuous beta belief function, it does not alter any other beta distribution when combined with. It represents full ignorance that the truth is in $[0, 1]$ but there is knowledge on whereabouts. Note that $\text{beta}(y; 0, 0)$ is not a beta distribution, which is undefined when $\alpha = \beta = 0$. It is semantically different from usual non-informative priors such as uniform prior $\beta(y; 1, 1)$ or Jeffreys prior (Jeffreys, 1961) $\text{beta}(y; 0.5, 0.5)$ or improper Haldane prior (Haldane, 1932), which all carries probabilistic knowledge on the truth, contrary to ignorance.

For another example, let $f_i(y) = ce^{-\frac{1}{2}yH_iy^T + \mu_iH_iy}$ be the density of Gaussian distribution $N(y; \mu_iH_i, H_i)$, where μ_i is the mean vector, H_i is the precision matrix or the inverse covariance matrix, μ_iH_i is precision-weighted mean vector, $i = 1, 2$, and c is a normalization constant. The product of the density functions is $e^{-\frac{1}{2}y(H_1+H_2)y^T + (\mu_1H_1 + \mu_2H_2)y} = e^{-\frac{1}{2}yHy^T + \mu Hy}$ up to normalization with precision $H = H_1 + H_2$ and precision-weighted mean vector $\mu H = (\mu_1H_1 + \mu_2H_2)$ or its normalized form $\mu = (\mu_1H_1 + \mu_2H_2)(H_1 + H_2)^{-1}$. Thus, per Dempster's rule, the combination of $N(y; \mu_iH_i, H_i)$, $i = 1, 2$, is Gaussian distribution $N(y; \mu_1H_1 + \mu_2H_2, H_1 + H_2)$. When y is univariate, The mean of the combined distribution is $\frac{H_1}{H_1+H_2}\mu_1 + \frac{H_2}{H_1+H_2}\mu_2$, and the variance $\frac{1}{H_1+H_2}$, which verifies the result in Morris (1977): the combination of two Gaussian distributions is also a Gaussian distribution with precision as the sum of the component distributions and mean as the precision weighted average of the component means. As a belief function, $N(y; 0, 0)$, with both zero precision and zero precision-weighted mean, is vacuous and has zero effect when combined with any other Gaussian belief functions. Again $N(y; 0, 0)$ is neither a Gaussian distribution nor a non-informative or improper prior. Like $N(y; 0, 0)$, additional constructs, such as observations, linear equations, conditional Gaussian distributions, etc., are included as special cases of Gaussian belief functions. In fact, Kalman filtering is verified as combining observations, linear equations, and Gaussian distributions as Gaussian belief functions (Liu, 2018). Gaussian belief functions can be also applied to combining probability judgments in financial asset pricing (Ayusuk and Sriboonchitta, 2015; Gao, 2016; Liu et al., 2006) and audit decision making (Srivastava and Liu, 2003).

3.2. Gamma belief functions

The formality is to extend the Dempster-Shafer theory to the infinite frame $\Theta = (0, +\infty)$ and a continuous variable of interest (Liu, 2024) with the mass density for $Y = y > 0$ being assigned as

$$f(y|\alpha, \beta) = \frac{\beta^\alpha y^{\alpha-1} e^{-\beta y}}{\Gamma(\alpha)} \quad (5)$$

where $\Gamma(\alpha)$ is the gamma function, and α and β are respectively shape and rate parameters. A gamma belief function represents epistemic knowledge on waiting time until α th arrival or spatial distance between α occurrences of Poisson events that occurs at rate β .

Assume $\Gamma(\alpha_1, \beta_1)$ and $\Gamma(\alpha_2, \beta_2)$ are two gamma belief functions on variable Y based on independent pieces of evidence. Their combination per Eq. (3) is

$$\frac{f(y|\alpha_1, \beta_1)f(y|\alpha_2, \beta_2)}{\int_0^\infty f(y|\alpha_1, \beta_1)f(y|\alpha_2, \beta_2)dy} = \frac{y^{\alpha_1+\alpha_2-2} e^{-(\beta_1+\beta_2)y}}{\int_0^\infty y^{\alpha_1+\alpha_2-2} e^{-(\beta_1+\beta_2)y} dy}.$$

Doing substitution $z = (\beta_1 + \beta_2)y$, it is easy to see

$$\begin{aligned} \int_0^\infty y^{\alpha_1+\alpha_2-2} e^{-(\beta_1+\beta_2)y} dy &= \left(\frac{1}{\beta_1+\beta_2}\right)^{\alpha_1+\alpha_2-1} \int_0^\infty z^{\alpha_1+\alpha_2-2} e^{-z} dz \\ &= \left(\frac{1}{\beta_1+\beta_2}\right)^{\alpha_1+\alpha_2-1} \Gamma(\alpha_1+\alpha_2-1) \end{aligned}$$

Thus, the combined gamma belief function becomes $f(y|\alpha, \beta) = \frac{(\beta_1+\beta_2)^{\alpha_1+\alpha_2-1} y^{\alpha_1+\alpha_2-2} e^{-(\beta_1+\beta_2)y}}{\Gamma(\alpha_1+\alpha_2-1)}$, which is also gamma. Thus, gamma belief functions are closed under Dempster's rule of combination.

The combination has shape parameter $\alpha = \alpha_1 + \alpha_2 - 1$ and rate parameter $\beta = \beta_1 + \beta_2$. As a special case, $\Gamma(\alpha, \beta) \oplus \Gamma(1, 0) = \Gamma(\alpha, \beta)$, implying that $\Gamma(1, 0)$ has a null effect when it is combined with $\Gamma(\alpha, \beta)$; it does not add nor remove information from $\Gamma(\alpha, \beta)$, and so semantically it is a vacuous belief function expressing full ignorance. Note that, $\Gamma(\alpha, \beta)$ is a Bayesian belief function when $\alpha, \beta > 0$. In contrast, $\Gamma(1, 0)$ is not a meaningful probability distribution because Eq. (5) is undefined. However, $\Gamma(1, 0)$ may be understood as the case when β approaches zero: $\Gamma(1, 0) = \lim_{\beta \rightarrow 0} \Gamma(1, \beta)$, or equivalently as the scale parameter $\theta = 1/\beta$ approaches infinity. A larger scale parameter means a larger uncertainty, and so a distribution with an infinite scale implies the complete uncertainty or full ignorance. Thus, $\Gamma(1, 0)$ is semantically a continuous extension to $m(\Theta) = 1$ that assigns the whole mass to $\Theta = (0, \infty)$ as the sole focal element. The same conception was implicit in Dempster (2001) and elaborated in Liu et al. (2006) for Gaussian belief functions.

One shall not confuse a continuous vacuous belief function with a uniform distribution. The former assigns the entire mass to the whole frame while the latter uniformly spread the mass over the frame. After all, there exists no uniform distributions over $\Theta = (0, \infty)$ for gamma variables or $\Theta = (-\infty, \infty)$ for Gaussian.

Therefore, we established the validity of gamma belief functions as a new family of continuous belief functions beyond Gaussian:

Definition 1. A gamma belief function, denoted by $\Gamma(\alpha, \beta)$, is a belief function on the strictly positive frame $\Theta = (0, +\infty)$: when there is probabilistic knowledge, a gamma belief function assigns belief mass to the frame using mass density $f(y|\alpha, \beta) = \frac{\beta^\alpha y^{\alpha-1} e^{-\beta y}}{\Gamma(\alpha)}$, and when there is ignorance, it assigns the entire mass to Θ as the sole focal element via the vacuous gamma belief function $\Gamma(1, 0)$.

Note that this concept currently models two special cases of knowledge, probabilistic knowledge as Bayesian belief functions and ignorance as vacuous belief functions. Intermediate cases with partial ignorance has not been studied yet.

Proposition 1. Assume $\Gamma(\alpha_1, \beta_1)$ and $\Gamma(\alpha_2, \beta_2)$ are two gamma belief functions encoding independence items of evidence. Then their combination via Dempster's rule is also gamma: $\Gamma(\alpha_1, \beta_1) \oplus \Gamma(\alpha_2, \beta_2) = \Gamma(\alpha_1 + \alpha_2 - 1, \beta_1 + \beta_2)$.

Note that the combination of gamma belief functions shall not be confused with the sum of gamma variables. Assume $Y_1 \sim \Gamma(\alpha_1, \beta)$ and $Y_2 \sim \Gamma(\alpha_2, \beta)$ are two independent gamma distributions with identical rates, then their sum is also a gamma $Y_1 + Y_2 \sim \Gamma(\alpha_1 + \alpha_2, \beta)$ (Mathai, 1982; Moschopoulos, 1985).

Let θ_1 and θ_2 be respectively the scale parameters of $\Gamma(\alpha_1, \beta_1)$ and $\Gamma(\alpha_2, \beta_2)$. Then, their combination has a scale parameter θ verifying $\frac{1}{\theta} = \frac{1}{\theta_1} + \frac{1}{\theta_2}$. Since $\theta_1, \theta_2 > 0$, $\frac{1}{\theta} > \max(\frac{1}{\theta_1}, \frac{1}{\theta_2})$, or alternatively, $\theta < \min(\theta_1, \theta_2)$, meaning the scale of $\Gamma(\alpha_1, \beta_1) \oplus \Gamma(\alpha_2, \beta_2)$ is smaller than the scale of both $\Gamma(\alpha_1, \beta_1)$ and $\Gamma(\alpha_2, \beta_2)$. Since scale parameter measures uncertainty, it makes an intuitive sense that the combined knowledge reduces uncertainty, which is the very purpose of combination.

As a probability distribution, $\Gamma(\alpha, \beta)$ has the mean $\mu = \alpha/\beta$ and the variance $\sigma^2 = \alpha/\beta^2$. Accordingly, the mean and variance of the combined gamma belief function are respectively

$$\mu = \frac{\alpha_1 + \alpha_2 - 1}{\beta_1 + \beta_2}, \sigma^2 = \frac{\alpha_1 + \alpha_2 - 1}{(\beta_1 + \beta_2)^2} \quad (6)$$

We will need this formula in the next section on combining predictions by gamma regressions. It is interesting to note that the combined variance, the scale parameter, of Gaussian belief functions has the same property as the scale parameter of gamma distributions.

In probability theory, gamma distributions constitute a unique family of parametric distributions, including exponential, Erlang, and χ^2 distributions as sub families: Erlang distribution $E_n(\lambda)$ is gamma $\Gamma(n, \lambda)$ with shape parameter as an integer n and rate parameter re-expressed as λ , exponential distribution $E(\lambda)$ is $\Gamma(1, \lambda)$ with rate parameter λ , and χ^2 distribution with degree of freedom n corresponds to $\Gamma(n/2, 2)$, a gamma distribution with rate $\beta = 2$. The Wishart distribution is a multivariate generalization of the gamma distribution. Applying Proposition 1 to these special cases, it is easy to see the following:

Corollary 1. Erlang and exponential are both closed families of parametric belief functions, but χ^2 is not. Exponential belief functions include exponential distributions $E(\lambda)$, when $\lambda > 0$, as Bayesian belief functions and $E(0)$ as a vacuous belief function. The combination of exponential belief functions $E(\lambda_1)$ and $E(\lambda_2)$ by Dempster's rule is also exponential: $E(\lambda_1) \oplus E(\lambda_2) = E(\lambda_1 + \lambda_2)$. Erlang belief functions include Erlang distributions $E_n(\lambda)$, when n is an integer and $\lambda > 0$, as Bayesian belief functions, and $E_1(0)$ as a vacuous Erlang belief functions. The combination of Erlang belief functions $E_n(\lambda_1)$ and $E_m(\lambda_2)$ by Dempster's rule is also Erlang: $E_n(\lambda_1) \oplus E_m(\lambda_2) = E_{n+m-1}(\lambda_1 + \lambda_2)$.

Note that $\chi_n^2 \oplus \chi_m^2 = \Gamma(\frac{n}{2} + \frac{m}{2} - 1, 4)$, which is not a χ^2 distribution. Thus, χ^2 distributions are not closed under combination and therefore do not make a valid family of parametric belief functions. However, exponential and Erlang mass densities both have the closure property, and thus they are distinct parametric sub families of gamma belief functions, establishing the validity of the notions of exponential and Erlang belief functions, which can model epistemic knowledge on time or distance between the occurrences of Poisson events. $E(0)$ and $E_1(0)$ are vacuous belief functions to model full ignorance in respective situations.

3.3. Normalized product intersection for fuzzy sets

While belief functions model ignorance or a lack of knowledge, fuzzy sets deal with vagueness or a lack of precision, which is common in verbal probability judgments (Mislavsky and Gaertig, 2022; Wallsten et al., 1997). Mathematically, a fuzzy set A of Θ is a pair $A = (\Theta, \mu_A)$ with μ_A as a membership function that maps each element $y \in \Theta$ to a degree of membership or possibility $\mu_A(y) \in [0, 1]$ (Zadeh, 1965, 1978). A fuzzy set is called normal if its height $H(A) = \sup_{y \in \Theta} \mu_A(y) = 1$. The classic notion of sets is a special case of fuzzy sets with 1 or 0 as memberships: $\mu_A(y) \in \{0, 1\}$ for any $y \in \Theta$.

Fuzzy set operations such as union and intersection can be defined through membership functions. Currently there are many definitions of such operations, all generalizing classic set operations but having different caveats. For intersections, minimum t-norm, $\mu_{A \cap B}(y) = \min\{\mu_A(y), \mu_B(y)\}$ (Zadeh, 1965) and algebraic product t-norm, $\mu_{A \cap B}(y) = \mu_A(y)\mu_B(y)$ (Schweizer and Sklar, 1963) are the most popular. The classic set intersection can be termed as either minimum or product t-norm of membership functions. For example, let $\Theta = \{y_1, y_2, y_3\}$ and $A = \{y_1, y_2\}$ and $B = \{y_2, y_3\}$ are two classic subsets. The two subsets may be expressed as fuzzy sets $A = \{1/y_1, 1/y_2, 0/y_3\}$ and $B = \{0/y_1, 1/y_2, 1/y_3\}$. Their intersection can be expressed as the product-intersection of memberships: $A \cap B = \{1 \times 0/y_1, 1 \times 1/y_2, 0 \times 1/y_3\}$.

Both minimum t-norm and product t-norm are commutative, associative, and monotonic, but other definitions may lack one or more such desirable features (Dombi, 1982; Dubois and Prade, 1980; Hájek, 1998; Klir and Yuan, 1995). The common problem of all definitions, however, is that applying to normalized fuzzy sets may lead to non-normalized intersections. Therefore, in order to maintain the normality of intersections, normalized t-norm is needed. The intersection of two fuzzy sets, A and B , may be defined as a function of their memberships in general:

$\mu_{A \cap B} = i(\mu_A, \mu_B)$, and its normalized form is, $\forall y \in \Theta$

$$\mu_{A \cap B}(y) = \frac{i(\mu_A(y), \mu_B(y))}{\sup_{y \in \Theta} i(\mu_A(y), \mu_B(y))} \quad (7)$$

It is shown that algebraic product t-norm is the only intersection operation that the normalized intersection is associative (Denceux, 2024; Denceux and Kreinovich, 2024).

Example 2. Let $A = \{0.8/y_1, 0.5/y_2, 0.1/y_3\}$ and $B = \{0.1/y_1, 0.9/y_2, 0.3/y_3\}$ be two fuzzy sets of $\Theta = \{y_1, y_2, y_3\}$. Using minimum t-norm, their intersection is

$$A \cap B = \{\min(0.8, 0.1)/y_1, \min(0.5, 0.9)/y_2, \min(0.1, 0.3)/y_3\} = \{0.1/y_1, 0.5/y_2, 0.1/y_3\}.$$

The normalized intersection is $A \cap B = \{0.2/y_1, 1.0/y_2, 0.2/y_3\}$ by using 0.5 as the normalization constant. According to the product t-norm, the intersection is

$$A \cap B = \{0.8 \times 0.1/y_1, 0.5 \times 0.9/y_2, 0.1 \times 0.3/y_3\} = \{0.08/y_1, 0.45/y_2, 0.03/y_3\}.$$

The normalized product intersection is $A \cap B = \{0.18/y_1, 1.00/y_2, 0.07/y_3\}$ per Eq. (7).

3.4. Gamma fuzzy sets

In literature, membership functions are often assumed to be in triangular or trapezoidal shape. Yet, in practice, these convenient shapes are not good approximations in the opinion of experts because they have more information on uncertainty, and alternatives such as Gaussian and lognormal shaped membership functions are proposed (Pacheco and Vellasco, 2009). For example, Gaussian fuzzy sets are used to represent solvent concentration and exposure duration to assess the health risk due to the exposure to chlorinated organic solvent (1, 1, 2-trichloroethene) via the consumption of contaminated drinking water (Dutta and Ali, 2012). The concept is further extended to Gaussian random fuzzy sets as a model that defines a Gaussian probability distribution over Gaussian fuzzy sets (Denceux, 2022, 2023).

Here I propose gamma fuzzy sets as yet another alternative that may be better to express uncertainties on quantities such as exposure duration (Dutta and Ali, 2012) or other time, money, and risk related judgments with imprecision. A gamma fuzzy set is a fuzzy set of $\Theta = \mathbb{R}^+ = (0, \infty)$ where the membership for $y \in \mathbb{R}^+$ is described by a gamma density function $f(y|\alpha, \beta)$ as in Eq. (5) with shape parameter α and rate parameter β with $\alpha, \beta > 0$. Let $\tilde{\Gamma}(\alpha, \beta)$ denote the fuzzy set whose membership is $f(y|\alpha, \beta)$ as in Eq. (5). A normalized gamma fuzzy set has a membership function, which is $f(y|\alpha, \beta)$ divided by the height of $\tilde{\Gamma}(\alpha, \beta)$:

$$H(\tilde{\Gamma}(\alpha, \beta)) = \sup_{y \in \mathbb{R}^+} f(y|\alpha, \beta)$$

Then, the normalized version of membership function is

$$\frac{f(y|\alpha, \beta)}{\sup_{y \in \mathbb{R}^+} f(y|\alpha, \beta)}$$

Lemma 1. When $\alpha \geq 1$, the gamma membership function in Eq. (5) has maximum at $y = \frac{\alpha-1}{\beta}$ and the maximum value is

$$H(\tilde{\Gamma}(\alpha, \beta)) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{\alpha-1}{\beta} \right)^{\alpha-1} e^{-(\alpha-1)}. \quad (8)$$

However, when $\alpha < 1$, $\lim_{x \rightarrow 0^+} f(y|\alpha, \beta) = +\infty$.

Proof. This lemma can be easily verified via the standard calculus. By taking the first-order derivative, we have

$$\begin{aligned} f'(y|\alpha, \beta) &= \frac{\beta^\alpha}{\Gamma(\alpha)} [(\alpha-1)y^{\alpha-2}e^{-\beta y} - \beta y^{\alpha-1}e^{-\beta y}] \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-2}e^{-\beta y}(\alpha-1-\beta y) \end{aligned}$$

Set $f'(y|\alpha, \beta) = 0$, we have $y = \frac{\alpha-1}{\beta}$ since $y > 0$ and $e^{-\beta y} > 0$. Taking the second-order derivative,

$$f''(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-3}e^{-\beta y}[(\alpha-1)(\alpha-2) - 2(\alpha-1)\beta y + \beta^2 y^2]$$

Plugging $y = \frac{\alpha-1}{\beta}$, we have

$$f''(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-3}e^{-\beta y}(1-\alpha).$$

Thus, $f''(y|\alpha, \beta) < 0$ if $\alpha > 1$, and so $y = \frac{\alpha-1}{\beta}$ achieves the maximum. Plugging $y = \frac{\alpha-1}{\beta}$ into Eq. (5), we arrive at Eq. (8).

When $\alpha = 1$, $f(y|\alpha, \beta) = \beta e^{-\beta y}$, which clearly has the maximum value β when $y = 0$ because $\beta > 0$. This is consistent with Eq. (8) by realizing $0^0 = \lim_{y \rightarrow 0^+} y^y = 1$.

When $\alpha < 1$, $f'(y|\alpha, \beta) < 0$ since $\alpha-1-\beta y < 0$. Thus, $f(y|\alpha, \beta)$ is a decreasing function, and its maximum can be reached when $y \rightarrow 0^+$:

$$\lim_{y \rightarrow 0^+} f(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lim_{y \rightarrow 0^+} \frac{1}{y^{1-\alpha}} = +\infty$$

□

This lemma precludes $\alpha < 1$ as proper shape parameters for gamma memberships because it implies infinite membership value for $y = 0$. Otherwise, a normalized gamma fuzzy set has membership function as

$$f(y|\alpha, \beta) = \left(\frac{\beta y}{\alpha-1} \right)^{\alpha-1} e^{-\beta y + \alpha-1} \quad (9)$$

and the height of a normalized gamma fuzzy set is 1.

Remark 1. A gamma distribution describes the time to Poisson events, and thus for survival analysis, the shape parameters should not be less than 1. However, when describing events with probabilities rapidly decreasing such as stock price drops after a drama events, early stage of a pandemic, etc., to use gamma fuzzy sets with such shape parameters, the data have to be left-censored.

Proposition 2. The normalized product intersection of two normalized gamma fuzzy sets, if their shape parameters are no less than 1, is also a gamma fuzzy set. In details, let $\tilde{\Gamma}_1(\alpha_1, \beta_1)$ and $\tilde{\Gamma}_2(\alpha_2, \beta_2)$ be two normalized gamma fuzzy sets. Then their normalized product-intersection is $\tilde{\Gamma}_1(\alpha_1, \beta_1) \odot \tilde{\Gamma}_2(\alpha_2, \beta_2) = \tilde{\Gamma}(\alpha_1 + \alpha_2 - 1, \beta_1 + \beta_2)$ if $\alpha_1, \alpha_2 \geq 1$.

Proof. Let $f_1(y|\alpha_1, \beta_1)$ and $f_2(y|\alpha_2, \beta_2)$ as in Eq. (9) be respectively the membership functions for two normalized gamma fuzzy sets $\tilde{\Gamma}_1(\alpha_1, \beta_1)$ and $\tilde{\Gamma}_2(\alpha_2, \beta_2)$ with $\alpha_1, \alpha_2 \geq 1$.

$$\begin{aligned} f_1(y|\alpha_1, \beta_1)f_2(y|\alpha_2, \beta_2) &= \left(\frac{\beta_1 y}{\alpha_1 - 1} \right)^{\alpha_1 - 1} e^{-\beta_1 y + \alpha_1 - 1} \left(\frac{\beta_2 y}{\alpha_2 - 1} \right)^{\alpha_2 - 1} e^{-\beta_2 y + \alpha_2 - 1} \\ &= \frac{y^{\alpha_1 + \alpha_2 - 2} e^{-(\beta_1 + \beta_2)y}}{\left(\frac{\alpha_1 - 1}{\beta_1} \right)^{\alpha_1 - 1} \left(\frac{\alpha_2 - 1}{\beta_2} \right)^{\alpha_2 - 1} e^{-(\alpha_1 + \alpha_2 - 2)}} \end{aligned}$$

As earlier, the maximum of this product is at $y = \frac{\alpha_1 + \alpha_2 - 2}{\beta_1 + \beta_2}$ with the maximum value as

$$H(\tilde{\Gamma}_1(\alpha_1, \beta_1) \cap \tilde{\Gamma}_2(\alpha_2, \beta_2)) = \frac{\left(\frac{\alpha_1 + \alpha_2 - 2}{\beta_1 + \beta_2} \right)^{\alpha_1 + \alpha_2 - 2}}{\left(\frac{\alpha_1 - 1}{\beta_1} \right)^{\alpha_1 - 1} \left(\frac{\alpha_2 - 1}{\beta_2} \right)^{\alpha_2 - 1}}$$

Therefore, the normalized product-intersection has a membership function as

$$\begin{aligned} \frac{f_1(y|\alpha_1, \beta_1)f_2(y|\alpha_2, \beta_2)}{H(\tilde{\Gamma}_1(\alpha_1, \beta_1) \cap \tilde{\Gamma}_2(\alpha_2, \beta_2))} &= \frac{y^{\alpha_1 + \alpha_2 - 2} e^{-(\beta_1 + \beta_2)y}}{\left(\frac{\alpha_1 + \alpha_2 - 2}{\beta_1 + \beta_2} \right)^{\alpha_1 + \alpha_2 - 2} e^{-(\alpha_1 + \alpha_2 - 2)}} \\ &= \left[\frac{(\beta_1 + \beta_2)y}{\alpha_1 + \alpha_2 - 2} \right]^{\alpha_1 + \alpha_2 - 2} e^{-(\beta_1 + \beta_2)y + (\alpha_1 + \alpha_2 - 2)} \end{aligned}$$

Per Eq. (9), this is the normalized membership function for $\tilde{\Gamma}(\alpha_1 + \alpha_2 - 1, \beta_1 + \beta_2)$. □

4. Combining gamma regressions

The gamma regression is a generalized linear model (McCullagh and Nelder, 1989) for predicting a variable of interest that follows a gamma distribution. All generalized linear models have three unifying

features: 1) the response variable follows a distribution in the exponential family (Andersen, 1970); 2) the linear combination of predictors determines the output; and 3) a link function $g(\cdot)$ relates the linear combination to the expected value μ of the distribution of the response variable as $g(\mu) = b_0 + \sum_{i=1}^n b_i X_i$ with $X_1, X_2, \dots, X_n$ being regressors and $b_0, b_1, \dots, b_n$ regression coefficients. Well known examples include the linear regression with the identity link for a response variable following a Gaussian distribution, the Poisson regression with the log-link for a response variables like counts or relative frequencies such as the number of claims, machine breakdowns, rainfalls, etc., the logistic regression with the logit link for a response variable following Bernoulli/binomial or categorical/multinomial distributions, and the Tweedie regression for a response variables such as total claim cost per policy holder, total interruption time per year, total rainfall per year, etc. following an inverse Gaussian distribution that has heavier tails than a gamma distribution.

Generalized linear models are estimated using the maximum likelihood principle. For gamma regression, observed responses $y_1, y_2, \dots$ has a log-likelihood as

$$\Sigma \log f(y_i | \alpha, \beta) = -\Sigma \beta y_i + \alpha \log \beta + (\alpha - 1) \Sigma \log y_i - n \log \Gamma(\alpha)$$

Reparameterizing the log-likelihood using mean μ and dispersion $\phi = 1/\alpha$ gives:

$$\Sigma \log f(y_i | \alpha, \beta) = \Sigma [t(y_i, \mu) / \phi + a(y_i, \phi)]$$

where $t(y_i, \mu) = -y_i / \mu - \log \mu$ and

$$a(y_i, \phi) = \frac{\log \phi}{\phi} + \left(\frac{1}{\phi} - 1\right) \log y_i - \log \Gamma\left(\frac{1}{\phi}\right)$$

In the classic gamma regression, assume the dispersion is invariant among observations, and the link function is $\log(\mu) = b_0 + \sum_{i=1}^n b_i X_i$. One chooses regression coefficients $b_0, b_1, \dots, b_n$ in $\mu = e^{b_0 + \sum_{i=1}^n b_i X_i}$ such that $\Sigma t(y_i, \mu)$ is maximized. The `glm()` function with the option, `family = Gamma(link = "log")`, in the R-core implements the computation.

A regression model is conceptually predictor value indexed conditional distributions of the response variable. As an example, using the data set of residential home values for Albemarle County, one can obtain a gamma regression model of Y (total home value) against the regressor X (finished area in square feet) as $\log(Y) = 11.78 + 5.081 \times 10^{-4} X$ with dispersion as $\phi = 0.2127$. This regression implies one conditional gamma distribution of Y given each value of X , e.g., when $X = 2000$, we have a gamma distribution $Y \sim \Gamma(\alpha, \beta)$ with $\alpha = 1/\phi = 4.711$, $\mu = e^{11.78 + 5.081 \times 10^{-4} \times 2000} = 361636.5$, and $\beta = \alpha/\mu = 1.303 \times 10^{-5}$, and when $X = 4000$, we have a different gamma distribution $Y \sim \Gamma(4.711, 4.716 \times 10^{-6})$. Note that, due the assumption of constant dispersion, the two distributions have identical shape parameters.

This leads to the idea of combining independent gamma regressions. The independence is justified if they are built in different studies as in meta analysis or in the same study but using different data partitions as in big data machine learning. Assume $\{\Gamma_1(\alpha_x^1, \beta_x^1) | x \in R^n\}$ and $\{\Gamma_2(\alpha_x^2, \beta_x^2) | x \in R^n\}$ are two independent gamma regressions on the same response variable Y against the vector X of n regressors. Then for each value of the regressors, say $X = x$, the two gamma distributions $\Gamma_1(\alpha_x^1, \beta_x^1)$ and $\Gamma_2(\alpha_x^2, \beta_x^2)$ are independent gamma belief functions or gamma fuzzy sets and can be combined into $\Gamma(\alpha_x^1 + \alpha_x^2 - 1, \beta_x^1 + \beta_x^2)$ per Propositions 1 or 2. Therefore, we obtain a new continuum of conditional gamma distributions, $\{\Gamma(\alpha_x^1 + \alpha_x^2 - 1, \beta_x^1 + \beta_x^2) | x \in R^n\}$, which represents the combination of the two gamma regressions. This is interesting; it not only allows independent gamma regressions to be combined as belief functions or fuzzy sets but also dictates such a combination is a closed operation. The closeness property entails that combination of three or more gamma regressions may be done sequentially. The combination of gamma belief functions and the intersection of fuzzy sets are commutative and associative, so is the combination of gamma regressions. Therefore, we have

Corollary 2. Assume $\{\Gamma_i(\alpha_x^i, \beta_x^i) | x \in R^n\}$, $i = 1, 2, \dots, m$, are independent gamma regressions against regressors X with values in R^n . Then their com-

bination via Dempster's rule or fuzzy set intersection is $\{\Gamma(\sum_{i=1}^m \alpha_x^i - m + 1, \sum_{i=1}^m \beta_x^i) | x \in R^n\}$, which is also a gamma regression.

Remark 2. When $\alpha_x^i < 1$, $i = 1, 2, \dots, m$, $x \in R^n$, normalized gamma fuzzy sets are undefined, but the conceptualization of gamma belief functions is intact, and thus Corollary 2 can be inferred by Proposition 1. The α parameter represents the total number of Poisson events, and so $\alpha \geq 1$ in normal practical situations. In contrast, $\alpha < 1$ means a fraction or incomplete Poisson event. It implies high probability of immediate occurrence with probability mass being mostly concentrated near zero. The situation may be seen in bursty or clustered arrival processes such as rare but explosive events, infant mortality, early failures, etc.

Regression models are meant for numerical prediction or classification. For gamma regressions, given regressor value $X = x$, the original component regression models predict the mean value of the response variable as α_x^i / β_x^i , $i = 1, 2, \dots, m$, while their combination predicts it as

$$\frac{\sum_{i=1}^m \alpha_x^i - m + 1}{\sum_{i=1}^m \beta_x^i}$$

Questions arise on the goodness of the combined prediction: how well does the combined model perform in comparison with each component regression model learned from a data partition and with the reference model built on the whole data, which is the merger of all partitions for building the component models? Note that the model based on the whole data serves as the reference for evaluation: It may not be available in practice since merging private data sets is not possible in meta analytics and not feasible in big data analytics. The upmost goal of any bootstrapping method is to create assemblies that perform as well as the reference model and outperform component models. Thus, the evaluation of goodness boils down to two different comparisons: 1) comparison between the predictions by the combined model and that by the reference model and 2) comparison among the predictions by the component models, the combined model, and the reference model against ground truth.

Remark 3. Although it is not studied in this paper, there may be another potential benefit of combining gamma regression models: resolving highly correlated predictors in gamma regressions. Note that, as in linear regressions, when predictors exhibit high multicollinearity, the covariance matrix of the maximum likelihood estimators can become inflated, leading to unstable or misleading coefficient estimates, including potential sign reversals (Akram et al., 2022; Lukman et al., 2022). This undermines the reliability of statistical inference, as standard errors increase and hypothesis tests lose power, making it difficult to assess the true effects of predictors. Aggregating gamma regression models trained on different data subsets may help mitigate the adverse effects of multicollinearity. Although multicollinearity may still exist within individual subsets, the aggregation process, by averaging over models with slightly varying correlation structures, can reduce the variance of coefficient estimates and produce more stable predictions. This averaging effect acts as a form of implicit regularization, potentially improving both predictive performance and robustness to collinearity-induced instabilities.

4.1. Experiments with classic gamma regressions

To compare the performance of combined model with that of the components and the reference model, in this subsection, I experiment with the combination of classic regressions based on real data on residential home values in Albemarle County. For illustration, using classic regression, we can first build component model m1: $\log(Y) = 11.39 + 6.393 \times 10^{-4} X$ with $\phi = 0.2558$ using homes 1 to 300, model m2: $\log(Y) = 11.85 + 4.637 \times 10^{-4} X$ with $\phi = 0.2135$ using homes 1000 to 1500, and then the reference model m: $\log(Y) = 11.39 + 6.393 \times 10^{-4} X$ with $\phi = 0.2558$ using the merged data consisting of homes 1–300 and 1000–1500. At $X = 2000$, the combined model predicts mean home value as 296749, and the reference model m predicts 316286, and

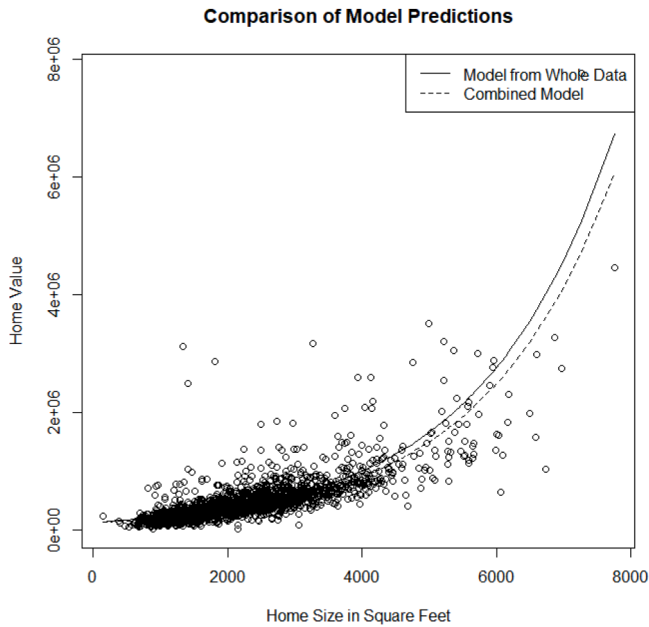


Fig. 2. Comparing combined gamma regression with reference model.

the relative difference, $|prediction(m) - prediction(mc)|/prediction(m)$, is 6.2%. At $X = 3000$, the combined model predicts mean home value as 511607 and the reference model m predicts 599405, and the difference is 14.6%. Note the difference gets bigger with increased home sizes. This is obvious as we can see from the scatter plot (see Fig. 2) that the reference model predicts well for smaller homes; after home size is over 3000, the prediction result is less reliable. This leads us to wonder which model predicts the actual home value better using error rate defined as $|actual - prediction|/actual$.

To counter the impact of sampling biases on error rates, I conducted a systematic experiment by splitting the whole data set into two partitions randomly using `createDataPartition()` function in “caret” package at split ratios = 20%, 30%, ..., 80% and at each split ratio running the splits five times to offset the impact of randomness. In total, I created 35 pairs of data partitions. For each pair, I built three gamma regression models: component model $m1$ from the first partition, component model $m2$ from the second partition, and the combined model mc by combining $m1$ and $m2$. In total, I had 35 pairs of component models and 35 combined models. All these models are compared with the reference gamma regression model m based on the whole data set. For a fair comparison, all component and reference models are learned under the same hyper parameter settings such as link functions, randomization, normalization, and regularization. Fig. 2 shows the prediction by the reference model m and the average predictions by 35 combined models. It shows that predictions by the combined model is very close to but almost always smaller than the prediction by the reference model m , and as home size increases, the difference gets bigger.

Table 1 quantifies the average relative differences along with 95% quantile ranges across all homes between models m and mc predictions. In details, I use the combined model and the reference model to predict the home values for each home in the sample, compute the absolute difference ratio $\frac{|prediction(m) - prediction(mc)|}{prediction(m)}$, and then take the average and 2.5% and 97.5% quantiles across all the homes. I do that in each of 35 runs with seven different split ratios and 5 repetitions. The table shows that the difference ratios have very tight 95% quantile ranges in all experimental runs, and the prediction made by the combined model is on the average about 10% away from that by the reference model. When the split ratio is 80% and home sizes of 7000 ~ 8000 square feet, the smallest differences is about 5 ~ 6% and the largest about 11 ~ 12%. At even splits at ratio around 40 ~ 50% and homes of sizes less than 3000

Table 1

Mean, minimum, and maximum differences in predictions between combined model and reference model.

Split	Run	Mean Diff	95%-Quantile Range
0.2	1	0.089	(0.078, 0.094)
0.3	1	0.105	(0.103, 0.105)
0.4	1	0.094	(0.094, 0.094)
0.5	1	0.104	(0.102, 0.104)
0.6	1	0.106	(0.106, 0.107)
0.7	1	0.087	(0.083, 0.094)
0.8	1	0.115	(0.107, 0.119)
0.2	2	0.095	(0.093, 0.099)
0.3	2	0.099	(0.096, 0.1)
0.4	2	0.108	(0.106, 0.109)
0.5	2	0.106	(0.104, 0.106)
0.6	2	0.091	(0.077, 0.097)
0.7	2	0.105	(0.095, 0.109)
0.8	2	0.094	(0.089, 0.104)
0.2	3	0.114	(0.113, 0.115)
0.3	3	0.103	(0.103, 0.104)
0.4	3	0.107	(0.107, 0.107)
0.5	3	0.103	(0.101, 0.108)
0.6	3	0.094	(0.093, 0.097)
0.7	3	0.109	(0.107, 0.114)
0.8	3	0.09	(0.085, 0.099)
0.2	4	0.093	(0.088, 0.095)
0.3	4	0.1	(0.094, 0.111)
0.4	4	0.103	(0.103, 0.103)
0.5	4	0.102	(0.092, 0.107)
0.6	4	0.104	(0.103, 0.105)
0.7	4	0.102	(0.102, 0.102)
0.8	4	0.088	(0.073, 0.096)
0.2	5	0.112	(0.11, 0.115)
0.3	5	0.089	(0.079, 0.093)
0.4	5	0.105	(0.105, 0.105)
0.5	5	0.101	(0.1, 0.104)
0.6	5	0.09	(0.086, 0.093)
0.7	5	0.096	(0.094, 0.1)
0.8	5	0.112	(0.109, 0.118)

square feet, we have smallest variations of differences, and data splits at ratios of 20 ~ 30% and 60 ~ 80% and home sizes greater than 5000, we have largest variations in the differences.

To evaluate the models against the actual home values, for each home, I use all four models: the reference model m , the combined model mc , and component models $m1$ and $m2$, to predict its value and compute the error ratios $|actual - prediction|/actual$. Then I compute the average error ratios across all homes and 95% quantile ranges of the errors. Table 2 shows the average prediction errors by mc and m . (The error rates by components $m1$ and $m2$ are not shown here since they are almost identical to the reference model m probably due to `createDataPartition()` function that ensures each subset is representative of the sample.) As we can see in the table, the combined model predicts better than the reference model. In all runs, the combined model has 95% of errors in the range between 1% and 100% with average error rate about 25% while the reference model has average 28% absolute error rates, and 95% quantile range between 1% and 130%. This result is somehow surprising, but it is clear the combined model does better than the reference model and each component model as well.

To understand the prediction errors further, I calculated signed prediction errors as $(actual - prediction)/actual$, which will have a sign indicating whether the prediction is over- or under-estimated. Then I calculated the average error rates across all the homes for all the four models. Table 3 shows that the reference model m on the average overestimated home values by 14%, and each component model does a bit worse at rates 13 ~ 18%, and all of them does a lot worse than the combined model mc , which fairly predicts the actual value on the average with error rate of 1 ~ 2%. This result is even more surprising, but it is consistent in all experiments. -0.14

Table 2
Prediction errors by the combined model and the reference model.

Split	Run	mc Mean Err	95%-Quantile Range	m Mean Err	95%-Quantile Range
0.2	1	0.257	(0.007, 1.06)	0.281	(0.007, 1.247)
0.3	1	0.256	(0.01, 1.02)	0.279	(0.007, 1.238)
0.4	1	0.256	(0.008, 1.045)	0.274	(0.008, 1.209)
0.5	1	0.256	(0.01, 1.022)	0.281	(0.007, 1.25)
0.6	1	0.255	(0.01, 1.016)	0.28	(0.007, 1.246)
0.7	1	0.257	(0.007, 1.06)	0.285	(0.007, 1.268)
0.8	1	0.255	(0.008, 0.999)	0.282	(0.006, 1.249)
0.2	2	0.256	(0.008, 1.042)	0.278	(0.007, 1.232)
0.3	2	0.256	(0.009, 1.033)	0.294	(0.009, 1.304)
0.4	2	0.255	(0.009, 1.012)	0.279	(0.008, 1.241)
0.5	2	0.255	(0.01, 1.018)	0.288	(0.006, 1.282)
0.6	2	0.257	(0.008, 1.057)	0.284	(0.006, 1.265)
0.7	2	0.256	(0.009, 1.022)	0.285	(0.006, 1.265)
0.8	2	0.256	(0.008, 1.044)	0.283	(0.007, 1.261)
0.2	3	0.255	(0.008, 0.998)	0.29	(0.006, 1.294)
0.3	3	0.256	(0.01, 1.023)	0.284	(0.006, 1.263)
0.4	3	0.255	(0.01, 1.015)	0.286	(0.006, 1.271)
0.5	3	0.256	(0.01, 1.023)	0.287	(0.006, 1.279)
0.6	3	0.256	(0.008, 1.043)	0.287	(0.005, 1.277)
0.7	3	0.255	(0.009, 1.009)	0.282	(0.006, 1.252)
0.8	3	0.257	(0.007, 1.053)	0.285	(0.006, 1.267)
0.2	4	0.256	(0.009, 1.05)	0.288	(0.006, 1.28)
0.3	4	0.256	(0.01, 1.031)	0.286	(0.007, 1.272)
0.4	4	0.256	(0.01, 1.023)	0.277	(0.008, 1.225)
0.5	4	0.256	(0.01, 1.029)	0.283	(0.007, 1.257)
0.6	4	0.256	(0.01, 1.02)	0.277	(0.007, 1.231)
0.7	4	0.256	(0.01, 1.025)	0.282	(0.007, 1.251)
0.8	4	0.257	(0.007, 1.063)	0.283	(0.006, 1.257)
0.2	5	0.255	(0.008, 1.004)	0.295	(0.008, 1.306)
0.3	5	0.257	(0.008, 1.059)	0.274	(0.008, 1.211)
0.4	5	0.256	(0.01, 1.019)	0.281	(0.007, 1.248)
0.5	5	0.256	(0.011, 1.028)	0.284	(0.007, 1.266)
0.6	5	0.257	(0.007, 1.054)	0.291	(0.007, 1.293)
0.7	5	0.256	(0.009, 1.039)	0.286	(0.006, 1.27)
0.8	5	0.255	(0.008, 1.004)	0.279	(0.007, 1.244)

4.2. Experiments with generalized gamma regressions

Classic gamma regression assumes invariant dispersion. [Cepeda-Cuervo \(2001\)](#) lifts the assumption and fits gamma regression models allowing both mean and shape parameters to follow regression structures. The algorithm is implemented in the “Gammareg” package for R ([Bossio and Cuervob, 2015](#)). [Corollary 2](#) does not have any restrictions on the structures, and it is worth an experimental valuation in case of varying dispersion as well.

Since the new regression algorithm permits log-link functions for the shape parameter α and either identity-link or log-link for the mean parameter μ ([Bossio and Cuervob, 2015](#)), so I conducted experiments with two different regression structures, in which the ground truth for shapes is $\log(\alpha) = 0.2 + 0.1X_1 + 0.3X_3$ in both structures, and that for the mean is respectively $\mu = 5 + 2X_1 + 3X_2$ and $\log(\mu) = -5 + 0.2X_1 - 0.03X_2$. As in [Bossio and Cuervob \(2015\)](#), simulated data sets were reproduced as follows: the uniform distributions $U(0,30)$, $U(0,15)$, and $U(10,20)$ are used to generate predictors X_1, X_2, X_3 , and gamma distribution $\Gamma(\alpha, \beta)$ is used to generate the response variable, where α and β are determined by the ground truth.

Within each regression structure, I repeatedly generate a pair of two data sets with each having respectively 500 and 800 observations and build regression models m1 and m2 using the two data sets and the reference model m by pooling the two data sets. Then I combine m1 and m2 into the combined model mc per [Corollary 2](#). Note that, instead of splitting one whole data set into various subsets as in the previous subsection, now each time I generated two simulated data subsets and merge them into a new whole data set.

This is an example elaborating how each experiment was conducted within the first regression structure that uses the log link

for the mean parameters. Two random data sets are generated using $\alpha = e^{0.2+0.1X_1+0.3X_3}$ for shapes and $\beta = \frac{\alpha}{\mu} = \frac{e^{0.2+0.1X_1+0.3X_3}}{e^{-5+0.2X_1-0.03X_2}}$ for rates, and from which the two component models are drawn respectively as m1: $\log(\alpha) = -0.349 + 0.1081X_1 + 0.3303X_3$ and $\log(\mu) = -4.999 + 0.1998X_1 - 0.02985X_2$, and m2: $\log(\alpha) = 0.3924 + 0.1009X_1 + 0.2845X_3$ and $\log(\mu) = -4.9998 + 0.2001X_1 - 0.02845X_2$. Then from the full dataset made of the two datasets, the reference model m is drawn as $\log(\alpha) = 0.1549 + 0.1028X_1 + 0.2995X_3$ and $\log(\mu) = -4.998 + 0.2000X_1 - 0.03025X_2$. We can see that the reference model m approximate well the ground truth: $\log(\alpha) = 0.2 + 0.1X_1 + 0.3X_3$ and $\log(\mu) = -5 + 0.2X_1 - 0.03X_2$, but none of the component models does. However, after combining these two poor component models, we can also see that the combined model does equally or even better than the reference model.

To perform the combination, for each sample, e.g., given $(X_1, X_2, X_3) = (20.37, 1.0447, 15.07)$, we compute shapes and rates as follows: $\alpha_1 = 926.62$, $\mu_1 = 0.382302$, and $\beta_1 = \alpha_1/\mu_1 = 2423.8$ per component model m1, $\alpha_2 = 843.10$, $\mu_2 = 0.384749$, and $\beta_2 = \alpha_2/\mu_2 = 2191.3$ per component model m2, and $\alpha = 864.72$, $\mu = 0.3839$, and $\beta = \alpha/\mu = 2248.47$ per the reference model m. According to [Proposition 1](#) and [2](#), the combined model mc will have shape $\alpha_C = 1768.72$ and rate $\beta_C = 4615.1$. The combined model has prediction $\mu_C = \alpha_C/\beta_C = 0.3832$, which is close to the predicted $\mu = 0.3839$ by the reference model m, closer than both component models. Note that the reference model m is built on the whole dataset, and the combination mc was indirectly inferred from the results based on partial datasets. Of course, one will not expect the combined model mc does any better than the reference model m. Yet it is desirable that the combined model predicts as closely as the reference, and both models performed equally well when evaluated against actual observations. To check these objectives, I calculate the relative

Table 3

Average prediction error rates of combined model, component models, and reference model.

Split	Run	mc Mean Err	m1 Mean Err	m2 Mean Err	m Mean Err
0.2	1	-0.039	-0.133	-0.142	-0.14
0.3	1	-0.021	-0.130	-0.144	-0.14
0.4	1	-0.033	-0.117	-0.156	-0.14
0.5	1	-0.022	-0.135	-0.145	-0.14
0.6	1	-0.019	-0.133	-0.15	-0.14
0.7	1	-0.041	-0.144	-0.13	-0.14
0.8	1	-0.009	-0.137	-0.152	-0.14
0.2	2	-0.032	-0.129	-0.143	-0.14
0.3	2	-0.027	-0.165	-0.129	-0.14
0.4	2	-0.017	-0.131	-0.146	-0.14
0.5	2	-0.020	-0.151	-0.13	-0.14
0.6	2	-0.037	-0.143	-0.137	-0.14
0.7	2	-0.021	-0.145	-0.129	-0.14
0.8	2	-0.033	-0.141	-0.137	-0.14
0.2	3	-0.010	-0.157	-0.136	-0.14
0.3	3	-0.022	-0.144	-0.138	-0.14
0.4	3	-0.018	-0.148	-0.135	-0.14
0.5	3	-0.023	-0.150	-0.130	-0.14
0.6	3	-0.033	-0.150	-0.125	-0.14
0.7	3	-0.016	-0.136	-0.148	-0.14
0.8	3	-0.038	-0.144	-0.123	-0.14
0.2	4	-0.035	-0.151	-0.137	-0.14
0.3	4	-0.027	-0.147	-0.137	-0.14
0.4	4	-0.023	-0.124	-0.151	-0.14
0.5	4	-0.024	-0.138	-0.143	-0.14
0.6	4	-0.021	-0.125	-0.163	-0.14
0.7	4	-0.023	-0.138	-0.146	-0.14
0.8	4	-0.04	-0.141	-0.139	-0.14
0.2	5	-0.013	-0.167	-0.133	-0.14
0.3	5	-0.039	-0.114	-0.151	-0.14
0.4	5	-0.02	-0.136	-0.143	-0.14
0.5	5	-0.025	-0.142	-0.138	-0.14
0.6	5	-0.037	-0.159	-0.112	-0.14
0.7	5	-0.030	-0.146	-0.126	-0.14
0.8	5	-0.013	-0.131	-0.177	-0.14

difference between the prediction by the reference model m and that by the combined model mc , which is $|\mu - \mu_C|/\mu = 0.00176$ in this example, and the relative prediction errors by both the reference model m and the combined model mc against the actual response value $Y = 0.356352$ are respectively $|Y - \mu_C|/Y = 0.0754736$ for the combined model mc and $|Y - \mu|/Y = 0.07737$ by the reference model m .

Therefore, for the particular sample in question, both models predict very closely with relative difference 0.176%, and the combined model even slightly outperforms the reference model against the actual observation. To evaluate the overall performance, I did the same calculation for all samples and found that the predictions by the combined model and by the reference regression had average relative difference of 0.18052% and 95% quantile range of (0.000375%, 0.89455%); the two models made almost identical predictions. The combined model had the average prediction error of 4.275% against actual response values, and the 95% quantile range is (0.1668%, 16.7531%), while the reference model had average prediction error of 4.2984% with 95% quantile range (0.1637%, 17.0212%). The prediction errors have indiscernible differences in distributions between the two models, and surprisingly the combined model has smaller prediction errors than the reference model most of times.

In experiments within the second regression structure based the identity-link for the mean parameters, datasets are produced using $\alpha = e^{0.2+0.1X_1+0.3X_3}$ and $\beta = \frac{\alpha}{\mu} = \frac{e^{0.2+0.1X_1+0.3X_3}}{5+2X_1+3X_2}$. I found a similar result that both reference and combined models made almost identical predictions. Here is the result of one experiment: the predictions by the combined model and the reference had average relative difference of 0.19585% with 95% quantile range of (0.0036%, 0.9650%). When comparing with the actual observations, the combined model had average prediction error of 4.5127% with 95% quantile range (0.1522%,

17.82%), and the reference had average error of 4.516% and 95% quantile range (0.1522%, 18.068%). The distributions of the errors between the models had no visual differences. For the total 1300 samples, the combined model outperformed the reference model in 600 in their predictions.

The above result was based on two single experiments, one for each regression structure. By replicating the experiments tens of thousands of times, each time four data sets, two for each regression structure, of sizes randomly generated by Poisson distributions. The results are stable, reproducing above examples.

In these replicated experiments, we were also interested in the number of occasions in which the combined model outperformed the reference model. When using log-link for the mean parameter, we did two simulations of 1000 and 2000 replications each. Combined model performed better in 530 replications in the first and 1043 in the second simulation. In the case of using the identity-link for the mean parameter, combined models did better in 580 of 1000 times in the first and 1200 of 2000 in the second. Thus, the combined model outperformed the reference model statistically. The winning ratios by the combined model had a much wider distributional spread when using the identity link for the mean parameter than that using the log link, but their distributions are both positively skewed; the combined models are visually shown to outperform the reference models overall. One-tail paired t tests on the hypotheses: H_0 : winning ratio = 0.5 and H_1 : winning ratio > 0.5 have p values $p = 2 \times 10^{-16}$ for both sets of replications, further confirming that the combined model statistically outperformed the reference model in both types of generalized gamma regressions.

5. Conclusion

This paper proposed a new approach to probability judgment aggregation via either combination of belief functions or intersection of fuzzy sets and justified the popular multiplicative aggregation rule without invoking contradictory normative axioms or twisting the symmetric combination of expert judgments as asymmetric, sequential Bayesian revisions. The approach avoids disguising judgmental priors as sampling information, introducing a middleman as the beholder of the consensus, or requiring the knowledge of non-existent middleman as prior, and it arrives at the identical result as in the natural-conjugate approach for combining beta probability judgments (Winkler, 1968), as in the likelihood function approach for combining Gaussian probabilities without invariance to scale and invariance to shift assumptions (Morris, 1977) or location-invariance assumption (Winkler, 1981), and as in the Kalman filter for combining dynamic measurements and predictions in automatic control and non-stationary time series analysis with added beauty that all evidence involved in the analysis, including equations, distributional assumptions, and observations, are all uniformly expressed as special cases of belief functions (Liu, 2018).

This paper contributes to the theory and application of belief functions and fuzzy sets in two significant ways. First, it demonstrates that gamma distributions are closed under Dempster's rule of combination (in the context of belief functions) and under the normalized product intersection (in the context of fuzzy sets). The result supports the formalization of gamma fuzzy sets and gamma belief functions, with the degenerate distribution $\Gamma(1, 0)$ representing a state of complete ignorance. Within the broader family of gamma belief functions, exponential and Erlang distributions form two closed subfamilies. In contrast, χ^2 distributions are not closed under combination and thus do not constitute a distinct subfamily.

Second, the paper establishes that the combination of gamma distributions extends naturally to the combination of gamma regression models. For classic gamma regression under the assumption of constant dispersion, the study conducts repeated random data splits of a real-world dataset. It finds that predictions from the combined model—formed by aggregating component models trained on partial data—differ by approximately 5% to 12% from those of a reference model trained on

the full dataset. For generalized gamma regression with varying dispersion, using simulated data, the combined model's predictions are nearly indistinguishable from those of the reference model, with differences ranging from 0% to 0.9%. Furthermore, when comparing model predictions against ground truth, the combined model consistently outperforms both the individual component models and the reference model in experiments involving classic gamma regression (on real data) and generalized gamma regression (on simulated data). These results are robust across 35 experiments on real data and over 2000 experiments on simulated data.

The results have an important implication to big data analytics where the volume, velocity, and variety of data often prohibit doing regressions on the complete data set. This paper paves an avenue to walk around the issue. For example, one may split data into multiple subsets, conduct gamma regression on each subset, and then combine the component regression models leading to a combined model approximating well the model based on the whole dataset. Since the combination of gamma regressions is also gamma, one may even do so dynamically: as more data come in, one can build a new model to update the existing one, instead of constantly merging new data and conducting new regressions based on ever growing datasets. Another use case of the results is meta analysis or quantitative integration of research findings. In academics, scholars often conduct regressions on similar variables of interest with different data, and one may want to integrate their findings to obtain an overall or unbiased understanding of the phenomenon, and yet he may not have access to all the data sets due to data privacy or unavailability. This paper provides a theoretical approach to the problem with no need of accessing and merging the data. Of course, care must be taken when interpreting aggregate models. While combined gamma regressions offer improved predictive accuracy and scalability for large or distributed datasets, they share an important limitation with all aggregated models. Specifically, the regression coefficients in aggregated models represent weighted contributions across multiple subsets of the data, rather than the marginal effect of any individual predictor. As such, although these models are highly effective for prediction and big data analytics, caution is warranted when interpreting coefficients for the purposes of decision-making or causal inference.

The gamma distributions belong to the exponential family of distributions, and it is tempting to inquire if the exponential family is also closed under the combination operation. Of course, the exponential family has a wide range of distributions with different parametric forms. It may not be sensible to combine arbitrary distributions in the family, e.g., to combine a discrete distribution with a continuous one. So far we know Gaussian, gamma, exponential, Erlang, beta, Dirichlet, categorical, and Bernoulli distributions are all closed under combination. It is important for future studies to examine if other sub family of distributions with the same parameters are also closed; a positive result will enable the aggregation of a wider range of probability judgments using the multiplicative formula or applying bootstrapping methods to a wider range of generalized linear models in data analytics.

Unlike other evidential ensemble method that assigns probability mass numbers to discrete predictions and then combine the mass numbers via Dempster's rule as support for combined prediction, this paper proposes ensemble learning of gamma regression models directly as gamma belief functions or gamma fuzzy sets. Future studies shall systematically review and compare these ensemble methods with respect to their theoretical support and empirical performance, and a particular target of comparison is the Gaussian evidential method that can directly combine models learned by maximizing likelihood estimates (Liu, 2025, 2026). The second direction of future research shall look into probability judgment aggregation and ensemble learning of gamma regressions with shape parameters $\alpha < 1$ in applications to predicting infant mortality, early failures of innovations, and rare explosive events such as pandemics, which are often constrained by data scarcity. The third direction shall investigate potential benefits of the proposed combination method in resolving the multicollinearity issue in gamma regressions. To sta-

bilize coefficient estimates and improve generalization, regularization techniques such as ridge-Liu regression or penalized likelihood estimation (Liu, 1993), as well as dimension reduction approaches like principal component regression or partial least squares, can be seamlessly integrated into the proposed aggregation framework. By applying these techniques within each component gamma regression model, the framework can reduce variance inflation and mitigate the adverse effects of multicollinearity. Aggregating such regularized or dimension-reduced models not only improves predictive performance but also enhances the robustness of parameter estimates, particularly in high-dimensional or correlated data environments.

CRedit authorship contribution statement

Liping Liu: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Writing - review & editing, Visualization, Investigation, Validation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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